

Parametric Survival Models

METIMMOX-1 Economic Evaluation

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1 Overview

This report documents the parametric survival models used in the economic evaluation. A single joint model is fitted for both endpoints:

Table 1: Economic survival formulas

Endpoint	Formula
Overall survival	$\text{Surv}(\text{OSwk}, \text{Death}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} * \text{Rx} + \text{tmb_braf} * \text{Rx}$
Progression-free survival	$\text{Surv}(\text{PFSwk}, \text{Progression}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} * \text{Rx} + \text{tmb_braf} * \text{Rx}$

2 Distribution Selection

Table 2: Overall survival distribution comparison

Distribution	AIC	BIC	delta_AIC	delta_BIC
weibull	666.98	686.96	0.00	0.00
weibullph	666.98	686.96	0.00	0.00
gamma	667.31	687.28	0.32	0.32
llogis	668.86	688.83	1.87	1.87
gengamma	668.97	691.17	1.99	4.21
lognormal	670.97	690.95	3.99	3.99
genf	670.98	695.39	3.99	8.43
gompertz	673.66	693.63	6.67	6.67
exponential	686.14	703.89	19.15	16.93

Table 3: Progression-free survival distribution comparison

Distribution	AIC	BIC	delta_AIC	delta_BIC
lognormal	489.86	509.84	0.00	0.00
gamma	491.48	511.46	1.62	1.62
gengamma	491.68	513.87	1.82	4.04
weibull	492.54	512.52	2.68	2.68
weibullph	492.54	512.52	2.68	2.68
llogis	492.85	512.82	2.99	2.99
genf	493.68	518.09	3.82	8.26
gompertz	496.55	516.52	6.69	6.69
exponential	501.16	518.92	11.30	9.08

3 Coefficients

Table 4: Best-fit and Weibull PH coefficient estimates

Outcome	Distribution	Parameter	Estimate	SE	95% CI
OS	weibull	shape	1.7474	0.1883	[1.4147, 2.1583]
OS	weibull	scale	158.0336	94.1040	[49.1910, 507.7067]
OS	weibull	Age	-0.0043	0.0088	[-0.0216, 0.0130]
OS	weibull	sex1	-0.1969	0.1556	[-0.5019, 0.1082]
OS	weibull	RxExperimental arm	-0.2573	0.2113	[-0.6715, 0.1569]
OS	weibull	crp1	0.4732	0.3132	[-0.1406, 1.0871]
OS	weibull	tmb_braf1	0.0589	0.2291	[-0.3901, 0.5078]
OS	weibull	RxExperimental arm:crp1	0.0566	0.3898	[-0.7074, 0.8206]
OS	weibull	RxExperimental arm:tmb_braf1	0.0729	0.3223	[-0.5587, 0.7046]
PFS	lognormal	meanlog	3.4923	0.8246	[1.8760, 5.1085]
PFS	lognormal	sdlog	0.8145	0.0821	[0.6685, 0.9923]
PFS	lognormal	Age	0.0034	0.0116	[-0.0193, 0.0262]
PFS	lognormal	sex1	-0.2408	0.2227	[-0.6773, 0.1957]
PFS	lognormal	RxExperimental arm	-0.6050	0.3379	[-1.2672, 0.0571]
PFS	lognormal	crp1	-0.1670	0.4864	[-1.1203, 0.7864]
PFS	lognormal	tmb_braf1	0.1297	0.3568	[-0.5697, 0.8290]
PFS	lognormal	RxExperimental arm:crp1	1.3589	0.5812	[0.2197, 2.4981]
PFS	lognormal	RxExperimental arm:tmb_braf1	0.2643	0.4648	[-0.6467, 1.1752]
OS	weibullph	shape	1.7474	0.1883	[1.4147, 2.1583]
OS	weibullph	scale	0.0001	0.0002	[0.0000, 0.0023]
OS	weibullph	Age	0.0075	0.0154	[-0.0227, 0.0378]
OS	weibullph	sex1	0.3440	0.2744	[-0.1938, 0.8818]
OS	weibullph	RxExperimental arm	0.4496	0.3718	[-0.2791, 1.1783]
OS	weibullph	crp1	-0.8269	0.5524	[-1.9097, 0.2559]
OS	weibullph	tmb_braf1	-0.1029	0.4000	[-0.8868, 0.6810]
OS	weibullph	RxExperimental arm:crp1	-0.0989	0.6809	[-1.4333, 1.2356]
OS	weibullph	RxExperimental arm:tmb_braf1	-0.1275	0.5628	[-1.2305, 0.9756]
PFS	weibullph	shape	1.4805	0.1634	[1.1925, 1.8381]
PFS	weibullph	scale	0.0064	0.0081	[0.0005, 0.0764]
PFS	weibullph	Age	-0.0138	0.0162	[-0.0456, 0.0181]
PFS	weibullph	sex1	0.2028	0.3074	[-0.3996, 0.8052]
PFS	weibullph	RxExperimental arm	0.8028	0.4292	[-0.0384, 1.6440]
PFS	weibullph	crp1	0.1376	0.6736	[-1.1827, 1.4579]
PFS	weibullph	tmb_braf1	-0.3116	0.4970	[-1.2857, 0.6624]
PFS	weibullph	RxExperimental arm:crp1	-1.3604	0.8209	[-2.9693, 0.2485]
PFS	weibullph	RxExperimental arm:tmb_braf1	-0.6307	0.6769	[-1.9574, 0.6959]

4 Strategy Survival Curves

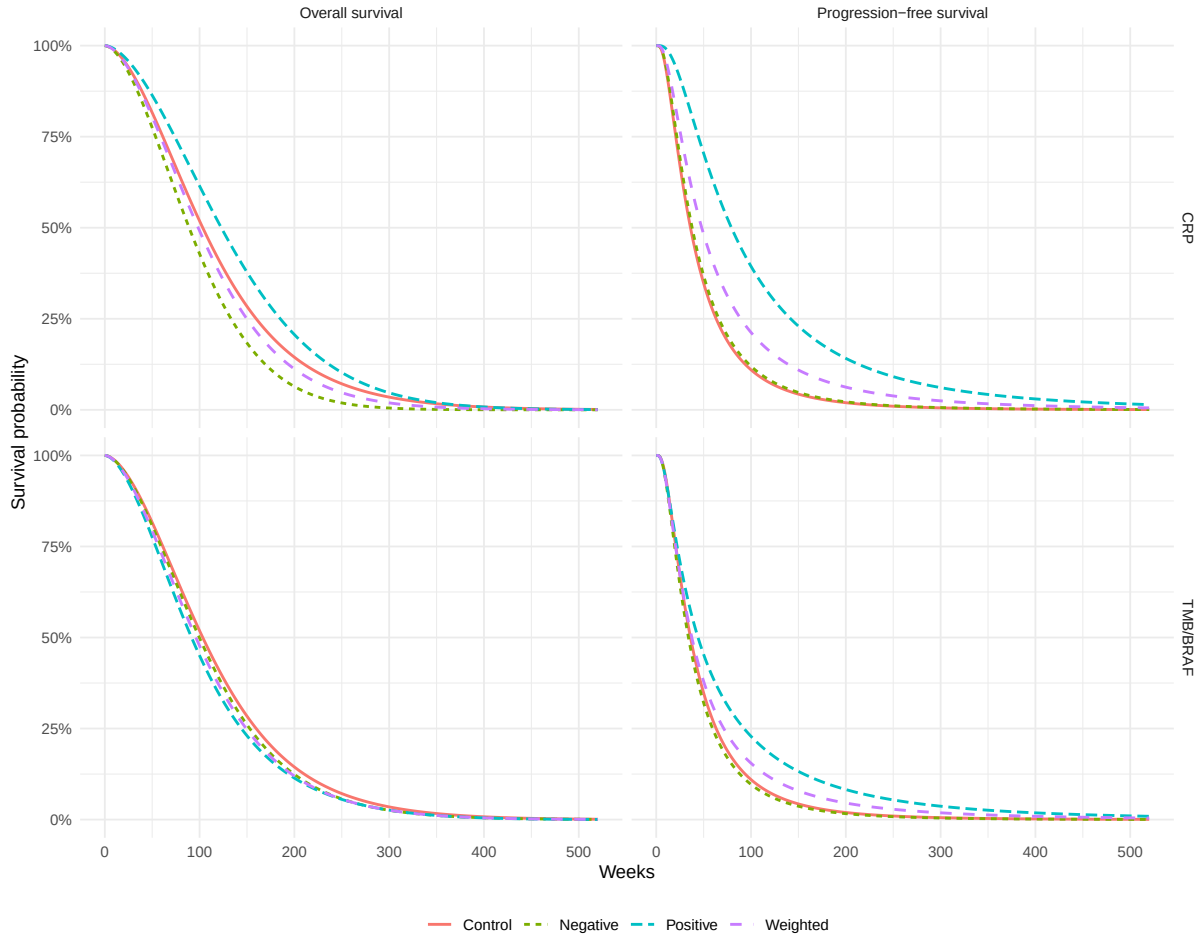


Figure 1: Predicted survival curves for economic biomarker strategies

5 Summary

The economic model uses one joint formula for both CRP-guided and TMB/BRAF-guided strategies. Parametric distribution choice is selected by AIC for each endpoint, with Weibull PH retained as a familiar reference fit.

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